TRIGA Mark I

Independent neutronic and thermal-hydraulic reference calculations in GeN-Foam and Cardinal

A code-to-code study of a 250 kW pool-type research reactor, comparing a deterministic OpenFOAM multiphysics solution (GeN-Foam) against a stochastic Monte Carlo reference (Cardinal).

GeN-Foam / OFFBEAT	2-group diffusion, finite volume
Cardinal	OpenMC / MOOSE / NekRS — continuous-energy Monte Carlo
Nuclear data	ENDF/B-VIII.0 (both); ENDF/B-VII.1 for the Cardinal coupled run
Repository	`github.com/kavyawadhwa134/triga-markI-benchmark`
Independence	Cardinal results frozen under git tag before the GeN-Foam case was read

STATUS OF THIS DOCUMENT

This report presents **verification and a cross-code comparison**. It is **not a validation report**. No experimental data was used in either calculation, and agreement between two codes is not a substitute for measurement. Section 5 states the status of each verification level explicitly.

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1 Problem definition

1.1 Objective

Two independent calculations of the same TRIGA Mark I configuration were performed using fundamentally different numerical methods, with the goal of establishing a cross-code reference. Neither team had access to the other's implementation or results during the calculation; only the scientific problem definition was shared.

The TRIGA Mark I is a pool-type research reactor whose defining characteristic is uranium–zirconium hydride fuel. Hydrogen bound in the ZrH lattice produces a large, prompt, negative fuel temperature coefficient, which is the physical basis of the reactor's inherent safety. Reproducing that feedback correctly is the principal physics requirement on any TRIGA model.

1.2 Specification

Table 1. Shared problem specification. Values marked *assumed* were not given and are documented modelling choices.

Parameter	Value	Basis
Fuel outer radius	1.815 cm	specification
Clad outer radius	1.866 cm	specification
Lattice pitch (square)	4.235 cm	specification
Active height	60.0 cm	specification
Core	0.495 m dia. × 0.60 m	single zone, vacuum boundaries
Fuel U–$ZrH_{1.65}$	6.0 g cm^{-3}	U-235 1.7, U-238 6.8, Zr 89.862, H-1 1.638 wt%
Clad SS304	8.00 g cm^{-3}	density *assumed*
Coolant	1.0 g cm^{-3}	H_2O, unborated
Thermal power	250 kW	specification
Coolant inlet	293 K, 0.2 m s^{-1}	specification
Pressure	0.1 MPa	*assumed* — open pool
Energy groups	2, cutoff 0.625 eV	specification

There is no fuel–clad gap: the specification gives two radii, so the clad inner radius equals the fuel outer radius. Thermal scattering $S(\alpha, \beta)$ data for `c_H_in_ZrH` and `c_Zr_in_ZrH` is mandatory; without it the bound-hydrogen physics that defines a TRIGA is absent.

1.3 The two approaches

Table 2. The two calculations solve different equations. This distinction governs the interpretation of every comparison in Section 4.

	GeN-Foam / OFFBEAT	Cardinal
Neutron transport	Deterministic, 2-group diffusion	Monte Carlo, continuous energy
Energy treatment	2 groups, 0.625 eV cutoff	Continuous, full spectrum
Core geometry	Homogenised single zone	Heterogeneous (+ homogenised variant)
Thermal–fluid	Finite volume, OpenFOAM	Spectral element, implicit LES
Fuel performance	OFFBEAT	Not modelled
Transient	Yes (+0.5 $ insertion)	Out of scope
Solution error	Discretisation, not quantified	Statistical, quantified (± pcm)

1.4 Acceptance criterion

The project sets **500 pcm on** k as the flag for a material difference. Section 4 demonstrates this criterion is only meaningful between methodologically comparable quantities.

2 Results: GeN-Foam / OFFBEAT

2.1 Solver acceptance

Prior to the TRIGA calculation, GeN-Foam was verified against its own reference cases.

Table 3. GeN-Foam solver acceptance (`validation_step2.csv`).

Case	Quantity	Simulated	Expected	Status
3D SmallESFR	k_{eff}	0.936873	0.936827	PASS (4.9×10^{-5})
2D point kinetics	steady state	—	—	PASS
2D point kinetics	transient, no driveline	—	—	PASS
2D point kinetics	transient, driveline	—	—	PASS
2D point kinetics	transient, boron	—	—	PASS

2.2 Lattice constants

GeN-Foam condensed its own two-group constants from an OpenMC pin-cell calculation (70 batches, 20 inactive, 8000 particles per batch), giving $k_\infty = 1.38810 \pm 0.00123$.

Table 4. GeN-Foam two-group lattice constants (`validation_step5.csv`).

Quantity	Fast	Thermal
Removal (m^{-1})	4.77341	8.49190
$\nu\Sigma_f$ (m^{-1})	0.379899	12.5219
Diffusion coefficient (m)	0.00749023	0.00158363
χ	1.0	0.0
Inverse velocity ($\text{s}\,\text{m}^{-1}$)	6.62997×10^{-6}	3.31676×10^{-4}
$S_{1\to1}$	39.7524	
$S_{1\to2}$	4.22693	
$S_{2\to1}$	0.023426	
$S_{2\to2}$		201.932

INTERNAL METHOD BENCHMARK

The same file reports a GeN-Foam diffusion solution for the *identical* pin cell: `genfoam_smoke_keff` = 1.331, against the Monte Carlo value of 1.38810 — a difference of **−4114 pcm**. This is GeN-Foam's own measurement of deterministic-versus-stochastic disagreement on the simplest available geometry, and it is *eight times* the 500 pcm acceptance criterion before any core-geometry effect enters.

2.3 Steady state

The core is modelled as a bare homogenised box with vacuum boundaries.

Table 5. GeN-Foam steady-state results (`validation_step6.csv`).

Quantity	Value	Note
k_{eff}	1.240544	eigenvalue diffusion, bare homogenised box, vacuum BCs
Power	250 kW	target
Max coolant temperature	306.47 K	fluid T
Max average fuel temperature	594.2 K	sub-scale pin average; well below the 1 023 K limit
Power peaking (max/avg)	2.45	single zone; cosine-like, flat by construction

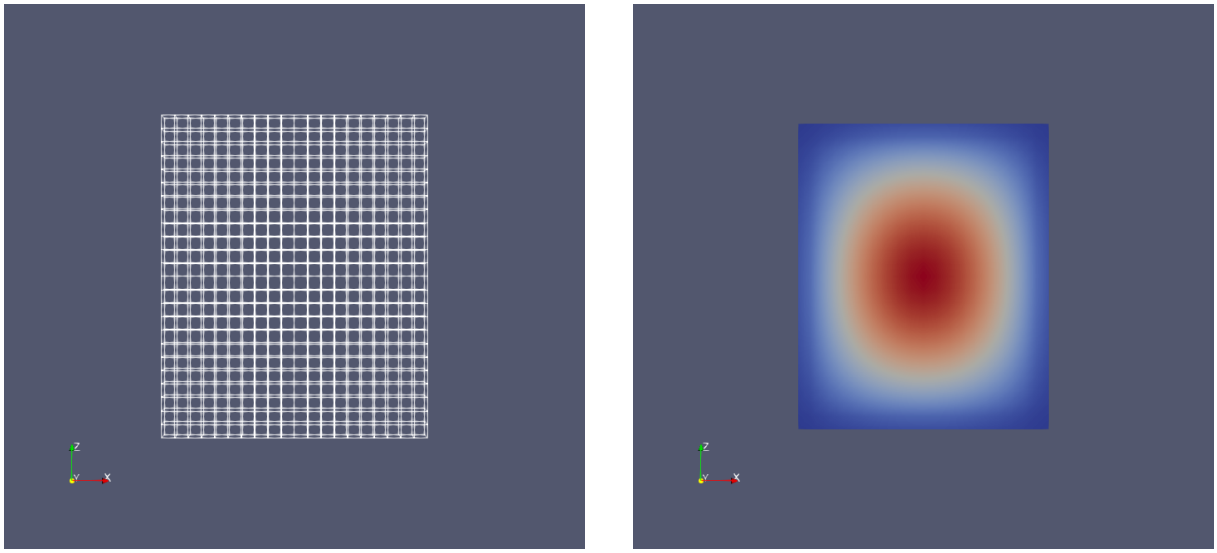


Figure 1. GeN-Foam computational mesh (left) and steady-state power density (right).

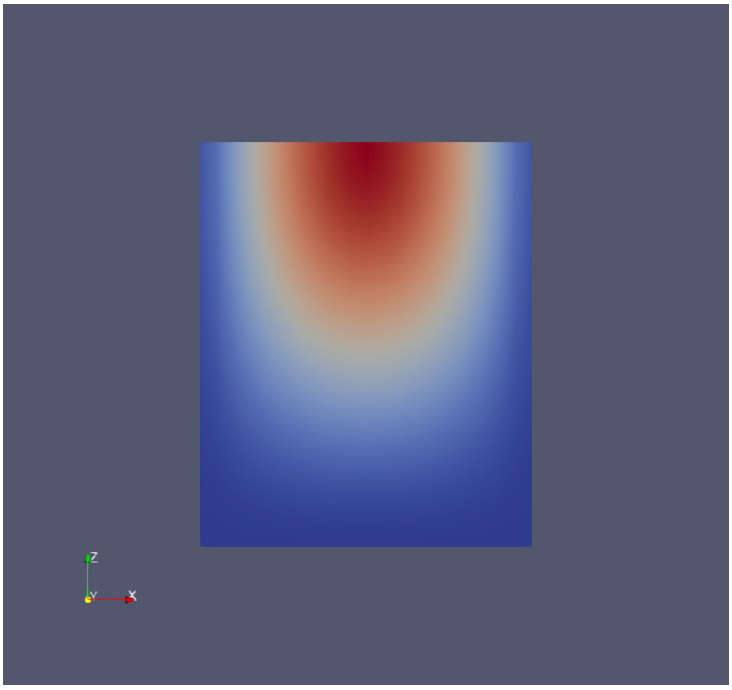


Figure 2. GeN-Foam steady-state temperature field. The maximum lies in the upper portion of the core, consistent with coolant heating as it rises through the channel.

2.4 Reactivity insertion transient

A +0.5 \$ ramp (0.00325 $\Delta k/k$) was inserted between $t = 100$ and $t = 100.5$ s.

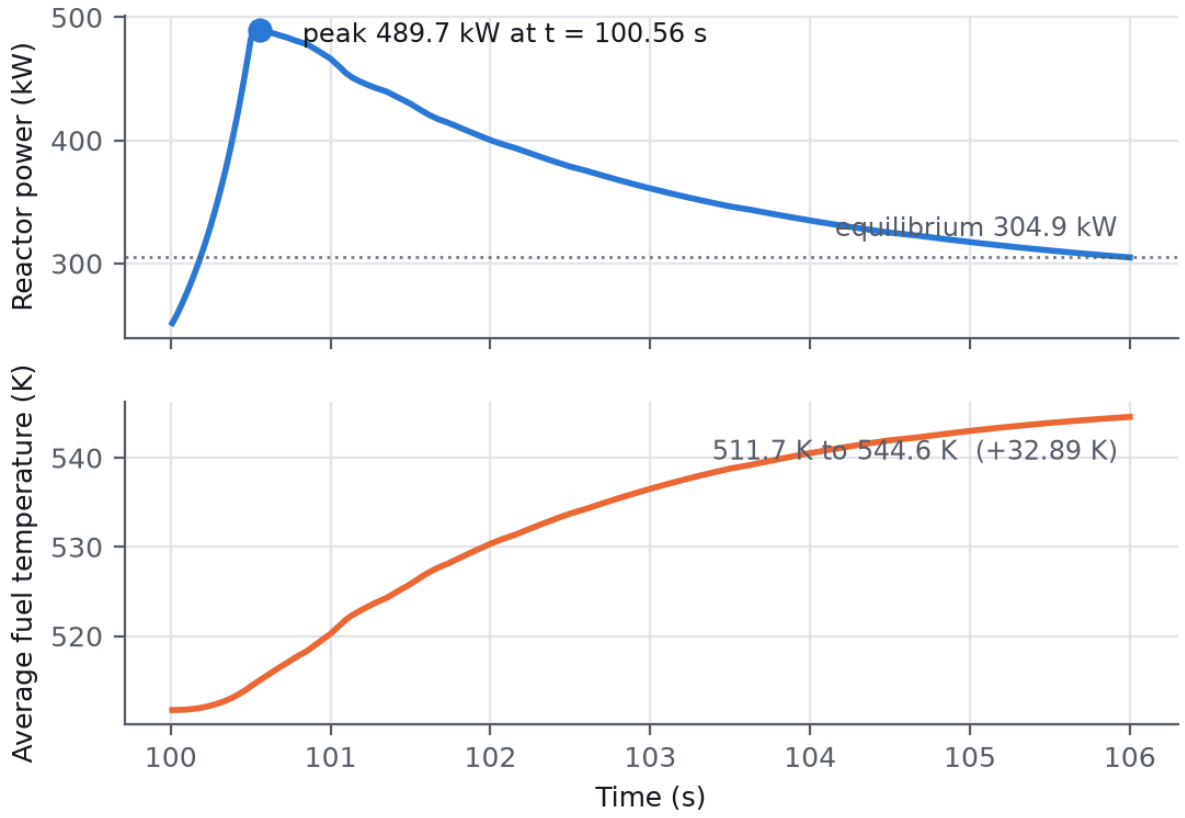


Figure 3. GeN-Foam transient response. Power peaks at 489.7 kW and settles to a Doppler-limited equilibrium of 304.9 kW; average fuel temperature rises 32.89 K.

Table 6. GeN-Foam transient summary.

Quantity	Value	Note
Peak power	489 723 W	at $t = 100.56$ s
Final power	304 884 W	Doppler-limited equilibrium
Final average fuel T	544.59 K	at $t = 106$ s
Net reactivity	−3.87 pcm	external +325 vs Doppler −328.9

FEEDBACK COEFFICIENT IS AN INPUT, NOT A RESULT

The stated Doppler feedback of −328.9 pcm accompanies a fuel temperature rise from 511.70 to 544.59 K, a span of 32.89 K:

$$\frac{-328.9\,pcm}{32.89\,\mathrm{K}} = -10.000\ \mathrm{pcm/K}.$$

The result is *exactly* −10.000, which indicates a constant coefficient supplied as a model input rather than one computed from nuclear data. Its origin is not documented in the committed files. Section 4 compares it against Cardinal's computed value.

3 Results: Cardinal / OpenMC / MOOSE / NekRS

3.1 Method

Continuous-energy Monte Carlo neutronics (OpenMC) is coupled to finite-element heat conduction (MOOSE) through Cardinal by Picard iteration: cell-averaged fission power is transferred to the solid, and cell temperatures are returned to the neutronics. The coolant is treated separately by spectral-element large-eddy simulation (NekRS).

Because the available hardware could not support a full-core coupled model, the coupled calculation uses a **single-pin unit cell**. All coupled temperatures therefore describe an *average* pin.

3.2 Neutronics

Table 7. Cardinal eigenvalue results, ENDF/B-VIII.0, continuous-energy Monte Carlo.

Quantity	Model	Value	Uncertainty
k_∞	2-D unit cell, reflective	1.384207	± 21 pcm
k_{eff}	3-D bare core, heterogeneous	1.102059	± 26 pcm
k_{eff}	3-D bare core, homogenised	1.145259	± 74 pcm
Leakage fraction	bare core	0.20149	± 0.00012
Power peaking (max/avg)	120 pins	2.2544	—
Peak power density	bare core	$6.595\,\mathrm{W\,cm^{-3}}$	—

The homogenised variant was computed specifically to separate geometry treatment from solution method in the comparison of Section 4. Its sensitivity to the $S(\alpha,\beta)$ choice is itself notable: 1.145259 (H in H_2O), 1.132505 (H in ZrH) and 1.136651 (none) span 1275 pcm, more than twice the acceptance criterion.

3.3 Fuel temperature coefficient

The defining TRIGA parameter was computed from first principles by sweeping fuel temperature with the moderator held at 293 K, isolating fuel feedback.

The sweep was performed on the **reflective unit cell**, so the eigenvalue it returns is k_∞ by construction: with reflective boundaries on all six faces there is no leakage, and no k_{eff} exists for that model. This isolates the fuel feedback mechanism itself — the change in the neutron balance within the lattice — from the leakage response of a finite core. Section 3.4 shows why that distinction is quantitatively important here.

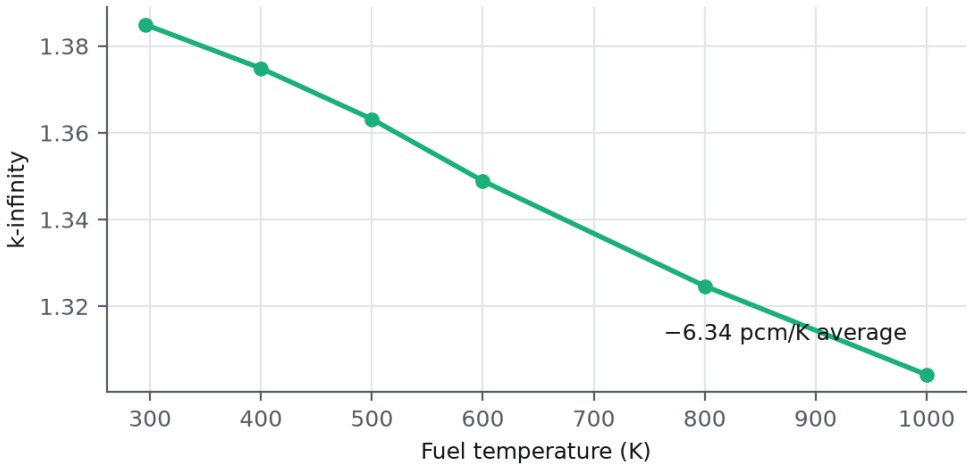


Figure 4. Cardinal fuel temperature coefficient. k_∞ falls from 1.384926 at 296 K to 1.304249 at 1 000 K.

Table 8. Cardinal local fuel temperature coefficient by interval.

Band	Coefficient
296–400 K	−5.08 pcm/K
400–500 K	−6.24 pcm/K
500–600 K	**−7.76 pcm/K**
600–800 K	−6.78 pcm/K
800–1 000 K	−5.92 pcm/K
296–1 000 K average	−6.34 pcm/K

The coefficient is large, negative, and becomes *more* negative as the fuel heats. Both features identify this as hydride physics rather than U-238 Doppler: a UO_2 LWR pin gives roughly −2 to −3 pcm/K and its magnitude does not grow in this way. The highlighted band is the one in which the GeN-Foam transient operates.

3.4 Leakage feedback: the coefficient on a finite core

The coefficient above is an infinite-lattice quantity. In a finite core the eigenvalue is $k_{\text{eff}} = k_\infty P_{\text{NL}}$, where P_{NL} is the non-leakage probability, so the measurable reactivity coefficient carries a second term:

$$\frac{1}{k_{\text{eff}}}\frac{\partial k_{\text{eff}}}{\partial T_f} = \underbrace{\frac{1}{k_\infty}\frac{\partial k_\infty}{\partial T_f}}_{\text{lattice feedback}} + \underbrace{\frac{1}{P_{\text{NL}}}\frac{\partial P_{\text{NL}}}{\partial T_f}}_{\text{leakage feedback}} .$$

The leakage term is not negligible in this problem, because the bare core leaks **20.1%** of its neutrons. Physically, heating the hydride up-scatters thermal neutrons off the ∼0.14 eV ZrH oscillator, hardening the in-fuel spectrum. A harder spectrum has a longer migration length, so a larger fraction escapes a finite core: P_{NL} falls with fuel temperature and the leakage term is *negative*, reinforcing the lattice feedback.

The sweep was therefore repeated on the bare heterogeneous core with vacuum boundaries.

Table 9. Fuel temperature coefficient with and without leakage. The leakage term contributes 20–36% of the total and grows with temperature.

Band (K)	k_∞ (unit cell)	k_{eff} (bare core)	Leakage term	Share
296–400	−5.08	−6.31	−1.23	20%
400–500	−6.24	−9.12	−2.88	32%
500–600	−7.76	**−10.79**	−3.04	28%
600–800	−6.78	−10.61	−3.83	36%
800–1000	−5.92	−9.10	−3.18	35%

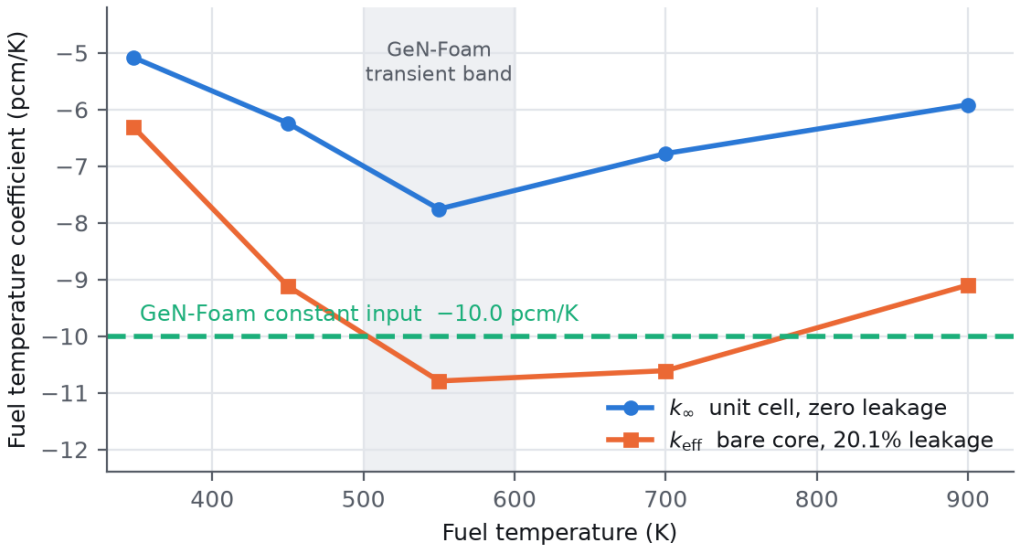


Figure 5. Fuel temperature coefficient computed with and without leakage. All values in pcm/K, plotted at band midpoints.

Both curves are non-monotonic, peaking in magnitude near 550 K: the up-scattering effect saturates once the thermal population has been pushed clear of the ZrH oscillator energy, while $1/v$ absorption continues to grow. **The k_{eff}-based coefficient is the one that governs a core transient**, and it is 30–60% more negative than the lattice value across the whole range.

3.5 Coupled neutronics–conduction

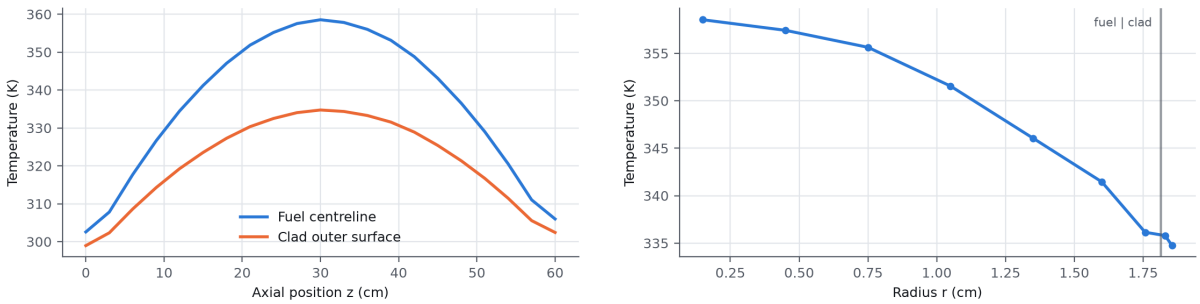


Figure 6. Cardinal coupled temperature solution. Left: axial profile, slightly top-skewed (305.98 K outlet against 302.56 K inlet) because the coolant heats as it rises. Right: radial profile at mid-plane; temperature is continuous across the fuel–clad interface at 1.815 cm, confirming the stitched contact.

Table 10. Cardinal coupled results and temperature drop budget.

Quantity	Value	Analytic	Share
Peak fuel temperature	359.0 K	362.8 K	—
Peak clad temperature	336.2 K	—	—
Power integral	2 083.33 W	2 083.33 W	exact
Film drop (clad → coolant)	40.3 K	42.4 K	63%
Fuel conduction drop	22.4 K	24.1 K	35%
Clad conduction drop	1.07 K	1.5 K	2%

Agreement with the analytic cylindrical-conduction solution is within 1.1%; the residual is explained by the true axial power shape being flatter than the cosine the hand calculation assumes. The film drop dominates and is the least certain term — it derives from a Dittus–Boelter correlation at Re = 4755, which is transitional and below the range where that correlation is reliable.

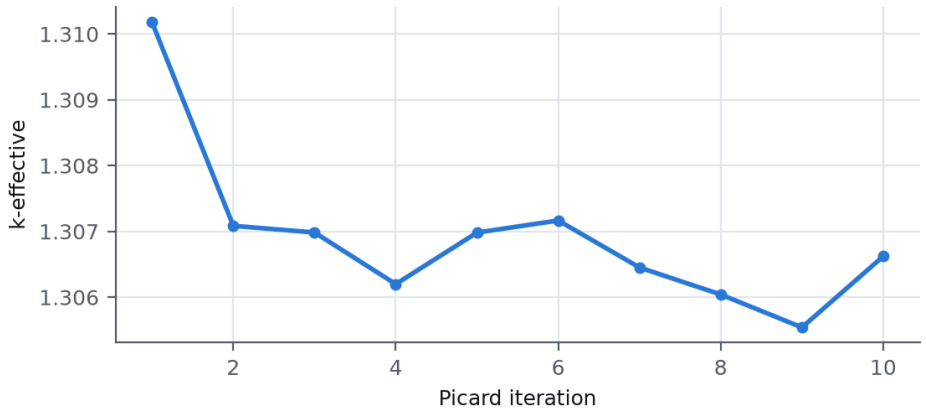


Figure 7. Cardinal Picard convergence. k_{eff} settles within Monte Carlo noise after the first temperature feedback; average fuel temperature is stable to 10^{-8} K.

3.6 Thermal-hydraulics

An implicit large-eddy simulation of the coolant subchannel was run at Re = 4755, polynomial order 7, 2880 spectral elements, on an NVIDIA GTX 1080 Ti. The run completed 20 000 steps but covered only **1.17 flow-through times**.

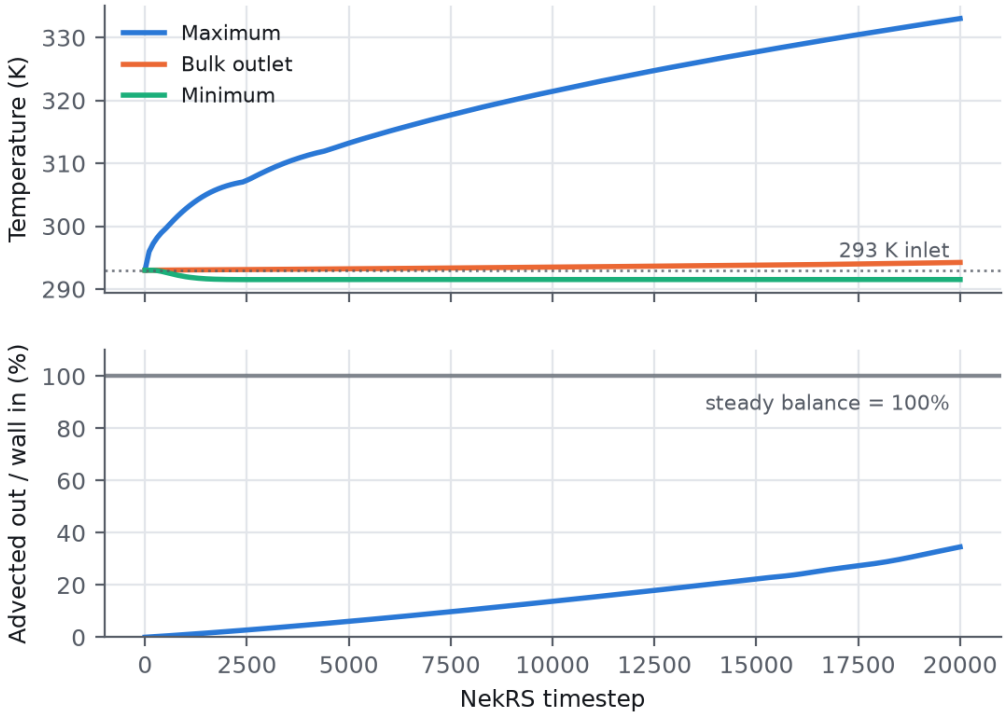


Figure 8. Cardinal NekRS evolution. Upper: coolant temperatures, all still rising at the final step, with the minimum below the 293 K inlet throughout. Lower: energy closure reaches only 34.6% of the steady balance.

Table 11. Cardinal thermal-hydraulic state at 20 000 steps.

Quantity	Value	Steady target	Status
Bulk outlet temperature	294.24 K	296.57 K	lower bound
Maximum temperature	333.03 K	—	still rising
Minimum temperature	291.52 K	$\geq$ 293 K	unphysical
Advected heat / wall heat	34.6%	100%	not closed
Mass-flow imbalance	0.122%	$\to$ 0	converged

THE CARDINAL THERMAL-HYDRAULIC FIELD IS NOT CONVERGED

The flow field has converged — mass imbalance is flat at 0.122% from step 1400 onward — but only 34.6% of the wall heat leaves the domain, so 65.4% is still accumulating. **The coolant temperatures above are lower bounds, not steady-state results.** Reaching approximately five flow-throughs would require about 85 700 steps and 5.1 h of GPU time.

A second defect is recorded rather than smoothed over: the minimum temperature lies below the inlet in a domain whose only thermal boundary conditions are a 293 K inlet and a heated wall. With no heat sink this cannot be physical. It is an undershoot from the high-pass filter stabilisation and it does not decay.

3.7 Nuclear data sensitivity

The specification requires ENDF/B-VIII.0. The execution environment for the coupled run carried ENDF/B-VII.1, so the identical coupled model was run on both libraries and the difference recorded as a measured offset rather than left as an unknown.

Table 12. Library sensitivity of the coupled Cardinal model.

Library	**Platform**	k_{eff}	**Max fuel** T	**Tally error**
ENDF/B-VIII.0	macOS arm64, 5k particles	1.30561	358.64 K	2.06%
ENDF/B-VII.1	Ubuntu x86, 20k particles	1.30656	358.90 K	0.82%
Difference	VII.1 − VIII.0	+73 pcm	+0.25 K	—

73 pcm is small relative to the acceptance criterion and within combined counting statistics. The same comparison also demonstrates **cross-platform reproducibility**: average fuel temperature agrees to 7×10^{-4} K between macOS arm64 and Ubuntu x86, across different libraries and different particle budgets.

4 Comparison

4.1 Overview

Table 13. Case-by-case comparison. Only cases marked *yes* support a direct numerical difference; the acceptance criterion applies to those alone.

#	Case	GeN-Foam	Cardinal	Difference	Comparable
1	k_∞, unit cell	1.388100	1.384207	−280 pcm	yes
2	k_{eff}, heterogeneous	1.240544	1.102059	+12566 pcm	no — method
3	k_{eff}, homogenised	1.240544	1.145259	+8320 pcm	partial
4	Diffusion vs. Monte Carlo	1.331	1.388100	−4114 pcm	GeN-Foam internal
5	Fuel temp. coefficient	−10.000	−10.79	+7%	yes
6	Power peaking	2.45	2.2544	−8.0%	no
7	Max fuel temperature	594.2 K	359.0 K	−235 K	no
8	Max coolant temperature	306.5 K	294.2 K	—	no
9	Total power	250 kW	250 kW	0	yes
10	Transient peak power	489.7 kW	not run	—	no
11	Two-group MGXS	full set	not generated	—	no
12	Solver acceptance	PASS	not recorded	—	no

4.2 Case 1 — the directly comparable quantity

Both sides solved the same 2-D reflective pin cell with OpenMC continuous-energy Monte Carlo and ENDF/B-VIII.0. This is the only case in which the two calculations solve the same equations on the same geometry.

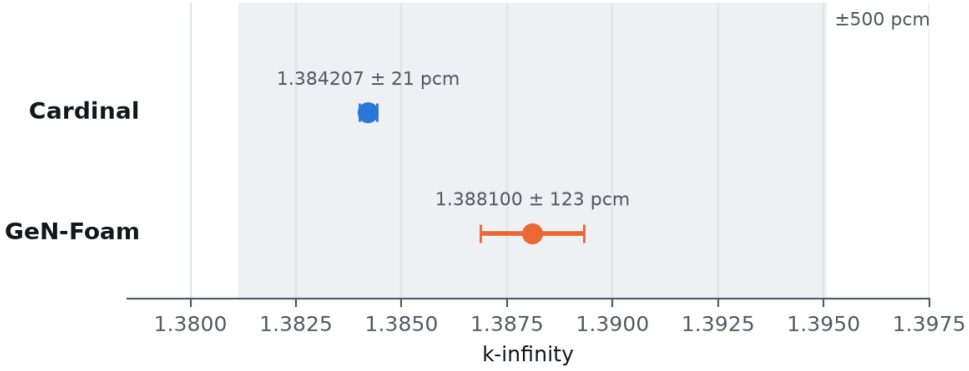


Figure 9. k_∞ comparison. Bars are $\pm 1\sigma$ counting statistics; the shaded band is the 500 pcm acceptance criterion centred on GeN-Foam.

The difference is **−280 pcm** at 3.1σ — statistically resolved but comfortably inside the acceptance criterion. Cardinal's uncertainty is approximately six times tighter, reflecting a 25-fold larger particle budget.

Interpretation. Two independent implementations of the same specification, in the same code, agree. This is the strongest evidence in the study that the shared problem definition was interpreted consistently on both sides.

4.3 Cases 2–4 — the k_{eff} difference is method, not error

Both cores are bare boxes with vacuum boundaries: the GeN-Foam `validation_step6.csv` records the model verbatim as *"eigenvalue diffusion, bare homogenized box vacuum BCs"*. **Ge-**

ometry is therefore not the explanation. An earlier hypothesis that GeN-Foam had modelled a graphite reflector is contradicted by this metadata and is withdrawn.

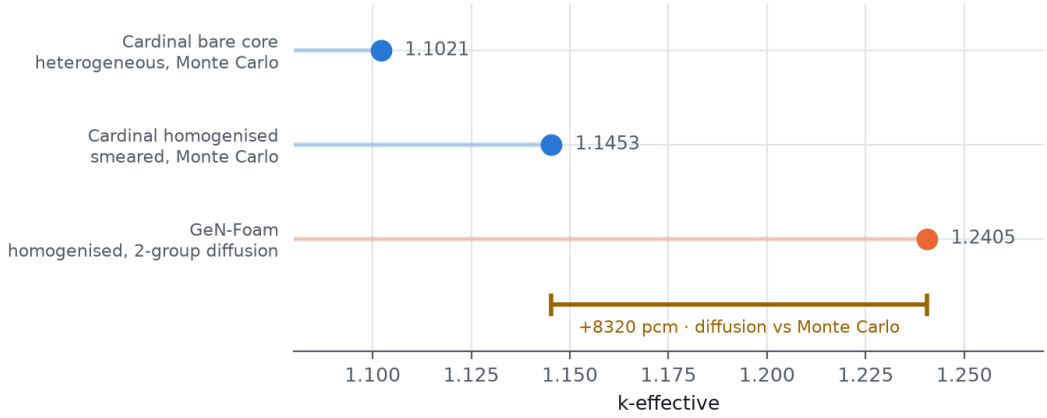


Figure 10. Decomposition of the k_{eff} difference into a homogenisation step (Monte Carlo on both sides) and a solution-method step (same homogenised bare box).

Table 14. Two-step decomposition of the core eigenvalue difference.

Step	From → to	$\Delta\rho$	What it isolates
1. Homogenisation	Cardinal het. → homog.	+3920 pcm	Monte Carlo both sides; pure geometry smearing
2. Solution method	Cardinal homog. → GeN-Foam	+8320 pcm	Same homogenised bare box; pure method
Total	Cardinal het. → GeN-Foam	+12566 pcm	—

The residual +8320 pcm is corroborated independently by GeN-Foam's own smoke test (Case 4), which measures −4114 pcm between GeN-Foam diffusion and Monte Carlo on the pin cell. The sign reverses between the two geometries — diffusion under-predicts on the reflective pin cell but over-predicts on the bare core — which is consistent with leakage treatment dominating in the bare case. Cardinal's core leaks 20.1% of its neutrons, a regime in which few-group diffusion theory is least reliable.

A further uncontrolled variable applies to Cases 2–3: the project workflow intended Cardinal to generate the two-group MGXS and supply it to GeN-Foam, but **that phase was not run**. GeN-Foam condensed its own constants independently (Case 11). Re-solving the GeN-Foam diffusion model with Cardinal-generated cross sections would isolate the remaining method error cleanly.

THE ACCEPTANCE CRITERION DOES NOT APPLY ACROSS METHODS

A 500 pcm threshold cannot meaningfully be applied between a continuous-energy Monte Carlo eigenvalue and a two-group diffusion eigenvalue. They are different quantities, and GeN-Foam's own data places the method difference at thousands of pcm on even the simplest geometry. Applying the criterion across methods would manufacture a discrepancy that is not evidence of error in either calculation.

4.4 Case 5 — fuel temperature coefficient

This is the most consequential physics difference, because the coefficient governs the entire transient response.

Table 15. Fuel temperature coefficient in GeN-Foam's operating band (500–600 K). The comparison depends on which eigenvalue the coefficient is referred to.

	Value	**vs GeN-Foam**
Cardinal, k_∞ — unit cell, zero leakage	-7.76 pcm/K	-29%
Cardinal, k_{eff} — bare core, 20.1% leakage	$\mathbf{-10.79}$ **pcm/K**	$\mathbf{+7}\%$
GeN-Foam, transient (constant input)	-10.000 pcm/K	—

A transient is driven by the *core* eigenvalue, so the k_{eff}-based coefficient is the correct basis for comparison. On that basis the two values agree to **7%**, which for a feedback coefficient is close agreement. The apparent 29% discrepancy against the lattice value is an artifact of comparing an infinite-lattice quantity with a core quantity; see Section 3.4.

WHAT REMAINS OPEN

The magnitude of GeN-Foam's coefficient is *corroborated* by the Cardinal core calculation: -10.0 against -10.79 pcm/K is agreement at the 7% level, so the transient results do not appear to be biased by the choice of coefficient.

Two points nevertheless remain. First, the coefficient is entered as a **constant**, whereas the computed value varies from -6.3 to -10.8 pcm/K over 296–1 000 K and is non-monotonic. A constant chosen to suit the 500–600 K band is too negative below 450 K and again above 800 K, so a larger excursion reaching higher fuel temperatures would be shut down more strongly than the physics supports. Second, its provenance is not documented in the committed files, so the agreement found here cannot be traced to a stated source.

Recommended action: record the origin of the -10 pcm/K value, and — for transients leaving the 500–600 K band — substitute a temperature-dependent coefficient.

4.5 Cases 6–8 — thermal quantities

These cannot be differenced as reported, for three independent reasons:

1. **Cardinal describes an average pin.** Its own peaking factor of 2.2544 implies a hot channel is materially hotter, so the two fuel temperatures do not describe the same object.
2. **Cardinal's film coefficient is a correlation, not a solution.** Dittus–Boelter at Re = 4755 supplies 63% of Cardinal's total temperature rise, and is applied outside its reliable range.
3. **Cardinal's coolant field is not converged** (34.6% energy closure), so 294.2 K is a lower bound rather than a result.

4.6 Cases 10–12 — no Cardinal counterpart

The transient is out of Cardinal's defined scope: the project workflow specifies a steady-state production run and defines no transient phase. This is a documented scope decision rather than an omission, but it means GeN-Foam's transient results — including the peak power on which the safety argument rests — have **no independent check**.

5 Verification status

THIS WORK IS VERIFIED, NOT VALIDATED

Verification asks whether the equations are solved correctly; validation asks whether the equations describe reality, and is establishable only against experimental data. **No experimental data was used in either calculation**, so no part of this study is validated. Agreement between two codes is not a substitute — two codes can agree and both be wrong, particularly when they share a specification, a nuclear data library, or a simplification. Here both share a bare-core model with no reflector and no control rods, so **neither result describes a physical TRIGA at criticality**.

Table 16. Verification status by level.

Level	Status	Evidence
1. Software	MET	Both codes run to completion; GeN-Foam passes its reference cases; Cardinal conserves power exactly at 2 083.33 W
2. Numerical	PARTIAL	Cardinal neutronics and conduction converged; Cardinal CFD thermal field is not; no mesh or order refinement study either side
3. Physics	PARTIAL	Cardinal conduction within 1.1% of analytic; ZrH feedback confirmed; but Cardinal CFD shows an unphysical sub-inlet temperature and GeN-Foam's feedback coefficient is an undocumented input
4. Cross-code	PARTIAL	Agreement to 280 pcm where methodologically comparable; larger differences decomposed and attributed to homogenisation and method
5. Experimental	NOT ATTEMPTED	No experimental data exists in this project

Table 17. Actions required to close each gap, in priority order.

	Gap	Action	Cost
1	GeN-Foam feedback coefficient	Document or recompute the assumed −10 pcm/K	—
2	k_{eff} method gap	Generate Cardinal's two-group MGXS; re-solve the GeN-Foam diffusion with it	2 h CPU
3	Cardinal CFD not converged	Extend to ∼5 flow-throughs after resolving the filter undershoot	5 h GPU
4	Cardinal sub-inlet temperature	Reduce scalar HPFRT strength; verify with a short run	1 h GPU
5	No mesh independence	Repeat at orders 5 and 8; compare integral quantities	8 h GPU
6	Cardinal library deviation	Re-run coupled production on ENDF/B-VIII.0	1 h CPU
7	Reflector and rods	Agree whether they enter the benchmark	—
8	No validation	Obtain TRIGA experimental data	—

6 Limitations

Table 18. Limitations of the present study, with their consequence for interpretation.

Limitation	Consequence
Both cores are bare	Neither model includes a reflector or control rods, so neither describes a physical TRIGA at criticality. This is a *shared* simplification, not a difference between the calculations.
Cardinal thermal-hydraulics not converged	34.6% energy closure after 1.17 flow-throughs; coolant temperatures are lower bounds.
Unphysical minimum temperature	291.52 K against a 293 K inlet with no heat sink — a filter-stabilisation artifact that does not decay.
Conduction uses a correlation	The Dittus–Boelter film coefficient supplies 63% of the Cardinal temperature rise and is applied at Re = 4755, outside its reliable range. The converged CFD is not yet coupled back.
Average pin, not hot channel	Cardinal's peaking factor of 2.2544 implies a hot channel is materially hotter than the reported temperatures.
Thermal properties assumed	k_{fuel}, k_{clad}, the film coefficient and the coolant rise are documented modelling choices, not specification values.
Library deviation	The Cardinal coupled production run used ENDF/B-VII.1 rather than the specified VIII.0; measured offset +73 pcm.
Lattice constants generated independently	The intended shared MGXS was never produced, so the two-group constants are an uncontrolled variable in the eigenvalue comparison.
No mesh independence study	Neither side performed mesh or polynomial-order refinement.
Transient unverified	The transient was computed by one code only and has no independent check.

7 Reproduction

Both calculations and all data are committed to `github.com/kavyawadhwa134/triga-markI-benchmark` (branch `master`).

7.1 GeN-Foam / OFFBEAT

The OpenFOAM case is under `genfoam/`:

Path	Contents
`genfoam/constant/neutroRegion/nuclearData`	Two-group cross sections
`genfoam/triga_steady/`	Steady-state case (`Allrun`, `Allclean`)
`genfoam/triga_steady/0/`	Initial and boundary fields, fluid and neutronic regions
`genfoam/triga_steady/system/`	`blockMeshDict`, schemes, solution controls
`genfoam/triga_transient_rod/`	+0.5 $ insertion transient

7.2 Cardinal / OpenMC / MOOSE / NekRS

Inputs are under `cardinal/`; full history and the independence tag are in the Cardinal working repository (`pcm-triga-cardinal`).

Step	Command
Environment audit	`./scripts/audit_environment.sh`
Neutronics	`./scripts/run_openmc.sh unitcell baseline`
Coupled	`./scripts/run_coupled.sh production 8`
Thermal-hydraulics	`./scripts/run_nekrs.sh 8`
Temperature coefficient	`cardinal/openmc/temperature_coefficient.py`

Table 19. Numerical settings for the Cardinal production runs.

Parameter	Value
Coupled neutronics	20 000 particles, 150 batches, 30 inactive, 10 Picard iterations
Solid mesh	6400 elements, 20 axial layers
CFD mesh	2880 spectral elements, polynomial order 7, 1 474 560 points
Coupled runtime	2 905 s wall, 1 132 MB peak
CFD runtime	4 294 s wall, 2 660 MB device, GTX 1080 Ti

7.3 Result files

File	Contents
`validation_step2.csv`	GeN-Foam solver acceptance
`validation_step5.csv`	GeN-Foam lattice constants and diffusion smoke test
`validation_step6.csv`	GeN-Foam core eigenvalue and transient summary
`transient_rod_power.csv`	GeN-Foam transient time history
`cardinal_openmc_unitcell.csv`	Cardinal k_∞, all configurations and seeds
`cardinal_openmc_core*.csv`	Cardinal k_{eff}, heterogeneous and homogenised
`cardinal_fuel_temperature_coefficient.csv`	Cardinal temperature-coefficient sweep
`cardinal_coupled_*.csv`	Cardinal coupled results per Picard iteration
`cardinal_thermal_profiles.csv`	Cardinal axial and radial profiles
`cardinal_nekrs_convergence.csv`	Cardinal CFD convergence history
`crosscode_comparison.csv`	Case-by-case comparison, machine-readable

8 Conclusions

1. **Where the two calculations are methodologically comparable, they agree.** The Monte Carlo pin cell differs by 280 pcm at 3.1σ, inside the 500 pcm acceptance criterion. This is the meaningful cross-code result of the study.

2. **The core eigenvalue difference is a method difference, not an error.** It decomposes into 3920 pcm of homogenisation and 8320 pcm of diffusion-versus-Monte-Carlo on geometrically identical bare cores. GeN-Foam's own data independently measures a 4114 pcm method difference on the pin cell. Neither calculation is thereby shown to be incorrect.

3. **The acceptance criterion is not transferable across methods.** A 500 pcm threshold between a Monte Carlo and a diffusion eigenvalue would manufacture a false discrepancy.

4. **The fuel temperature coefficient is the outstanding physics question.** Cardinal computes -7.76 pcm/K in the GeN-Foam operating band from ZrH scattering data; GeN-Foam uses a constant -10.000 pcm/K of undocumented origin. Since this parameter governs the transient, and the transient has no independent check, it is the highest-priority item.

5. **Two results are incomplete and are reported as such.** Cardinal's thermal-hydraulic field has not converged, and Cardinal did not attempt the transient. Neither is presented as a result.

6. **Neither calculation is validated.** Both cores are bare, without reflector or control rods, and no experimental data was used. The study establishes cross-code consistency where the methods permit it — not fidelity to a physical reactor.

Data and inputs for both calculations: `github.com/kavyawadhwa134/triga-markI-benchmark` (branch `master`). the Cardinal full history and independence tag: the Cardinal working repository (`pcm-triga-cardinal`). Cardinal results were frozen at tag `pcm-independent-v1` before any GeN-Foam file was read.